FOROUGH SHIRIN ABKENAR

Ph.D.

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PROFESSIONAL SUMMARY

ML Engineer and Researcher with expertise in **deep learning**, **generative AI**, and **domain adaptation**. Experienced in developing **cutting-edge AI models** and **fine-tuning LLMs** for **failure prediction**, **optimization**, and **decision-making** in dynamic environments. Strong programming skills in **Python** and **PyTorch**, with a deep understanding of **probabilistic modeling**, **representation learning**, and **AI-driven automation**. Passionate about advancing state-of-the-art ML techniques and applying them to real-world challenges.

PROFESSIONAL BACKGROUND

Postdoctoral Scholar

University of California Davis

February 2023 – February 2025 | Davis, CA

- Designed **deep learning models** for **failure prediction** and **anomaly detection**, improving predictive accuracy.
- Enhanced **model generalization** with **domain adaptation** (DANN, self-supervised) and improved efficiency via **knowledge distillation**.

Postdoctoral Scholar

University of California Irvine

February 2022 – February 2023 | Irvine, CA

- Collaborated with **Intel AI/ML teams** to develop **adaptive machine learning models** for time-variant datasets and real-time data processing, improving performance in **real-world AI applications**.
- Led **feature engineering efforts** and **statistical analysis** for large-scale datasets, optimizing data-driven insights.

Other Professional Activities

February 2025 – Present

- Pattern recognition and text classification using **few-shot prompting**, **prompt engineering**, and **LLM fine-tuning** (causal and sequential).

EDUCATION

Ph.D. in Electrical and Information Engineering

The University of Sydney

Mar 2018 – Jan 2022

Thesis title: Towards Hyper-efficient IoT Networks Using Fog Paradigm

M.Sc. in IT Engineering

Sahand University of Technology

Sept 2012 – June 2014

B.Sc. in IT Engineering

Shahid Madani University of Azarbayjan

Sept 2008 – June 2012

SKILLS

CNN | GAN | VAE | LLM | FL | RL

Python | PyTorch | Scikit-learn | C++ | SQL | MATLAB | GitHub | Linux

Deep Learning | Transfer Learning | Statistical Analysis | Feature Engineering | Optimization | Probabilistic Modeling

ACHIEVEMENTS

- **Generative AI with LLMs Certification**, Amazon Web Services, 2025.
- **Dean's Award for the Best Thesis**, The University of Sydney, 2023.

SELECTED PUBLICATIONS

Forough Shirin Abkenar, et al.:

Full list (20 papers with 450+ citations): GoogleScholar

Journal Articles

- "A Survey on Mobility of Edge Computing Networks in IoT: State-of-the-Art, Architectures, and Challenges," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2329–2365, 2022.
- "ENERGENT: An Energy-Efficient UAV-Assisted Fog-IoT Framework for Disaster Management," *Journal of Communications and Networks*, vol. 24, no. 6, pp. 698–709, 2022.

Conference Proceedings

- "Online Domain Adaptive Classification for Mobile-to-Edge Computing," in *IEEE WoWMoM*, Boston, MA, 2023.
- "Selective Data Offloading in Edge Computing for Two-Tier Classification With Local Domain Partitions," in *IEEE PerCom Workshops: Cloud2Things*, Atlanta, GA, 2023.